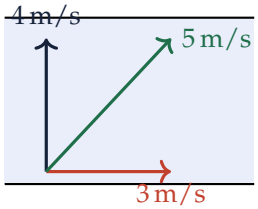
AP PHYSICS 1

Unit 1 • Lesson 1.4 • Notes (filled in)

Topic 1.4 • Reference Frames and Relative Motion

Notes (filled in)

Objective: describe motion from different reference frames and add relative velocities.

Reference frames	Position and velocity depend on the observer: the reference frame. A passenger sitting in a moving bus is at rest relative to the bus and moving relative to the road. Two frames moving at constant velocity relative to each other measure the same acceleration.
Adding relative velocities	$\vec{v}_{A \text{ rel } C} =$ $\vec{v}_{A \text{ rel } B} + \vec{v}_{B \text{ rel } C}$ A train moves at 20 m/s east. A passenger walks at 1.5 m/s relative to the train. Walking east: 21.5 m/s east relative to the ground. Walking west: 18.5 m/s east relative to the ground.
Example: crossing a river	A boat points straight across a river 80 m wide at 4 m/s relative to the water. The water flows at 3 m/s. Speed relative to the bank: $\sqrt{4^2 + 3^2} = 5$ m/s Time to cross: $80/4 = 20$ s (only the across velocity matters) Drift downstream: $3 \times 20 = 60$ m 
Check	Car A drives at 25 m/s east, car B at 15 m/s east. Velocity of A relative to B: 10 m/s east. If B drives west at 15 m/s: 40 m/s east.
Summary	Motion is always described relative to a reference frame. Velocities add as vectors: $\vec{v}_{A/C} = \vec{v}_{A/B} + \vec{v}_{B/C}$. Frames moving at constant velocity relative to each other agree on accelerations.